MIL-C-13902A
26 December 1956
SUPERSEDING
MIL-C-13902(CmlC)
23 December 1954

MILITARY SPECIFICATION

# COATING COMPOUND, ACETYLENE TETRACHLORIDE RESISTANT

*This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.*

## 1. SCOPE

1.1 This specification covers a baked type of protective coating for spray application which is resistant to the action of acetylene tetrachloride. The coating is intended primarily for use on the interior of containers.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

### SPECIFICATIONS

FEDERAL

TT-P-141 — Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

PPP-C-96 — Cans, Metal, 28 Gage and Lighter.

PPP-D-729 — Drums: Metal, 55-Gallon (For Shipment of Noncorrosive Materials).

PPP-D-760 — Drums and Pails, Metal (5 and 16.64 Gallon).

PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof — For Packaging and Sealing.

MILITARY

MIL-A-101 — Adhesive, Water-Resistant, For Sealing Fiberboard Boxes.

JAN-P-108 — Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN-T-247 — Tetrachlorethane (Acetylene Tetrachloride).

### STANDARDS

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MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

FSC 8010

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM FREIGHT CLASSIFICATION

Rating, Rules and Regulations

(Copies may be obtained from the Official Classification, 1 Park Avenue at 33d Street, New York 16, N. Y.)

## 3. REQUIREMENTS

### 3.1 Composition.

3.1.1 *Total solids.* The total nonvolatile content shall be not under 60.0 percent by weight of the coating compound when tested as specified in 4.4.1.

3.1.2 *Solvent.* The solvent shall be completely volatile and shall be not over 40.0 percent by weight of the coating compound when tested as specified in 4.4.1.

3.1.3 *Toxicity.* The material shall be free of benzol, carbon tetrachloride, or other substances of comparable toxicity.

### 3.2 Physical properties.

3.2.1 *Weight per gallon.* One gallon of the material shall weigh not less than 10.5 pounds when tested as specified in 4.4.1. The unit of purchase shall be 1 U. S. gallon at 23° ± 0.5° C.

3.2.2 *Flexibility.* The baked coating compound shall not crack when tested as specified in 4.4.3.

3.2.3 *Odor.* The odor of the package material shall be inoffensive.

3.2.4 *Spraying properties.* The package material when reduced as recommended by

3.2.5 *Acetylene tetrachloride resistance.* The baked film shall show no peeling, blistering, checking, or softening when tested as specified in 4.4.4.

3.2.6 *Dilution stability.* The compound shall show no separation, curdling, or precipitation when tested as specified in 4.4.5.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 **Lot.** A lot shall consist of the coating compound from one batch offered for delivery at one time (see 6.2).

4.2 **Sampling.** Unless otherwise specified (see 4.3.3) sampling shall be in accordance with method 102.1 of Specification TT–P–141.

4.2.1 *Sampling for inspection of filled containers at the contractor's plant.* A random sample of filled containers shall be taken by the inspector in accordance with Standard MIL–STD–105 at inspection level I and acceptable quality level equalling 2.5 percent defective to verify compliance with this specification in regard to fill, closure, marking, labeling and other requirements not involving tests.

4.2.2 *Sampling for tests at a Government laboratory.*

4.2.2.1 The inspector shall take two containers at random from each inspection lot (4.1). One quart representative specimens shall be taken from each of the two containers and placed in separate clean, dry, metal or glass containers and then sealed, marked, and forwarded to the test laboratory designated by the activity concerned except where delivery is to be made to Mare Island Naval Shipyard, Vallejo, California, or to Norfolk Naval Shipyard, Portsmouth, Virginia, samples shall be forwarded to the activity receiving delivery.

4.2.2.2 When requested by the contractor, the inspector shall take specimens in accordance with 4.2.2.1 with the following exceptions: Three-quart specimens shall be taken instead of 1-quart specimens. Each well



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the manufacturer and applied by spraying shall show no sagging, creeping, or running and be satisfactory in all other respects when tested as specified in 4.4.3.2.
mixed 3-quart specimen shall be divided into three equal parts, one part to be delivered to the contractor, one part to the testing laboratory, and one part to be held by the inspector in case of dispute.

4.2.2.3 *Testing activity.* Unless otherwise specified in the contract or order, the samples taken by the inspector as required for test purposes shall be forwarded to the Industrial Test Laboratory, Philadelphia Naval Shipyard 12, Pennsylvania; to the Industrial Laboratory, Mare Island Naval Shipyard, Vallejo, California; to the Chemical Laboratory, Norfolk Naval Shipyard, Portsmouth, Virginia; or to the Material Laboratory, New York Naval Shipyard, Brooklyn 1, New York (see 6.2).

**4.3 Inspection.**

4.3.1 Unless otherwise specified (see 4.5) inspection shall be in accordance with method 101.1 of Specification TT–P–141 except that the following additional requirements shall apply.

4.3.1.1 *Inspection of filled containers at the contractor's plant.* Each sample filled container taken as specified in 4.2.1 shall be examined for construction defects of the container and the closure, for leakage, and for unsatisfactory markings. Each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or that is under the required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming product.

4.3.2 *Lot acceptance tests.* The specimens taken in accordance with 4.2.2.1 shall be subjected to the tests specified in 4.4 except that the laboratory may perform complete tests on only one specimen from each lot provided sufficient tests are made on the second specimen to verify that the material represented by the two specimens is the same. Lots shall be accepted or rejected by the inspector on the transmitted specimens. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.3.3 *Alternative inspection.* When specified (see 6.1) inspection shall be made in accordance with section 1000 of Specification TT–P–141.

4.3.4 *Preparation for delivery.* The inspector shall ascertain that each lot complies with section 5.

**4.4 Test procedures.** The material shall be tested in accordance with applicable methods of Specification TT–P–141 and as specified herein.

4.4.1 The following tests shall be conducted in accordance with Specification TT–P–141:

| *Test* | *Method* |
|---|---|
| Total solids | 404.1 |
| Solvent | 404.1 |
| Weight per gallon | 401.1 |

4.4.2 *Toxicity.* The manufacturer shall submit a certificate that the material is free from benzol, carbon tetrachloride, or other toxic solvents.

4.4.3 *Flexibility.*
4.4.3.1 *Panel preparation.* Test panels shall be prepared as specified in method 201.1 of Specification TT–P–141.

4.4.3.2 *Coating application.* The material, thinned with the solvent recommended by the manufacturer, shall be sprayed on the test panels (4.4.3.1) air dried for 1 hour at ambient temperature and humidity, and baked in accordance with the manufacturers recommendation. The baked film shall be 0.004 ± .001 inch thick. The panels shall be observed during spraying and after baking for compliance with 3.2.4.



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4.4.3.3 Remove the baked panel from the oven, cool to 25° ± 5° C. and bend 180° over a ⅛-inch diameter metal mandrel within 1 minute at 25° ± 5° C. Observe bent portion for defects in coating material.

4.4.4 *Acetylene tetrachloride resistance.* Prepare, bake, and cool 3- by 5-inch panels as specified in 4.4.3. Immerse the panels in acetylene tetrachloride at 93° ± 5° C. for 100 hours. Approximately half of the 5-inch dimension shall be immersed in the test liquid which shall conform to JAN–T–247. Remove the panels from the test liquid, air dry for humidity and examine for conformance with 3.2.5.

4.4.5 *Dilution stability.* Dilute the package material with an equal volume of thinner recommended by the manufacturer as specified in method 439.1 of Specification TT–P–141. Mix thoroughly and observe for compliance with 3.2.6.

**4.5 Rejection and Resubmission.** If any specimen selected as specified in 4.2 fails to conform to this specification, the lot represented shall be rejected. The contractor, at no expense to the Government, shall have the option of having an analysis made on each container in the lot, removing the nonconforming material, and resubmitting the remaining portion of the lot for acceptance testing. If any specimen taken from the resubmitted portion of the lot fails to conform to this specification, the lot shall be finally rejected.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging.**

5.1.1 *Level A.* The material shall be packaged in a one gallon can conforming to type V, class 2 of Specification PPP–C–96. The exterior coating of the can shall conform to Plan B as specified thereto.

**5.2 Packing.**

5.2.1 *Level A.*

5.2.1.1 *One gallon cans.* Four one gallon cans of the material packaged as specified in 5.1.1 shall be packed in a style RSC, $V3_c$ or $V3_s$ box conforming to Specification JAN–P–108. The cans shall be floored 2 x 2. The flaps of the box shall be sealed with adhesive conforming to Specification MIL–A–101. The corners, joints, and seams of the box shall be sealed with type II, class 1, color optional tape conforming to Specification PPP–T–60. The box shall be strapped in accordance with appendix I of Specification JAN–P–108.

5.2.1.2 *Five-gallon pails.* The material shall be packed in a type II, class 2 pail conforming to Specification PPP–D–760.

5.2.1.3 *Fifty-five gallon drums.* The material shall be packed in a type III drum conforming to Specification PPP–D–729. The exterior coating of the drum shall be as specified thereto.

5.2.2 *Level C.* The coating compound shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

**5.3 Marking.** In addition to any special marking required by the contract or order, all unit packages and shipping containers shall be marked in accordance with Standard MIL–STD–129. Each container shall bear a label giving adequate instructions for solvent reduction, application, and baking schedule.

## 6. NOTES

**6.1 Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing.
(c) Testing laboratory to which specimens for test are to be sent.
(d) When inspection is to be conducted as specified in 4.3.3.



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6.2 **Batch.** A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever: and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

**Custodian:**
Army—Chemical Corps
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**Other interest:**
Army—E

**Preparing activity:**
Army—Chemical Corps



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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